



riscv-vela

Collaborative Open-Source System Software Project for RISC-V
Enabling an Open Software-Defined Robotics Ecosystem

Vela — inspired by the constellation of the Sails, symbolizing a new voyage



Project Overview

Open Architecture
RISC-V Standards for Design
Freedom Without Vendor Lock-In

Open-Source Development
Community-Driven Development
Through GitHub

Open SDR Ecosystem
Royalty-Free System Software for
Software-Defined Robotics

Core Design Goals: *Low Power · High Performance · High Reliability · High Availability*

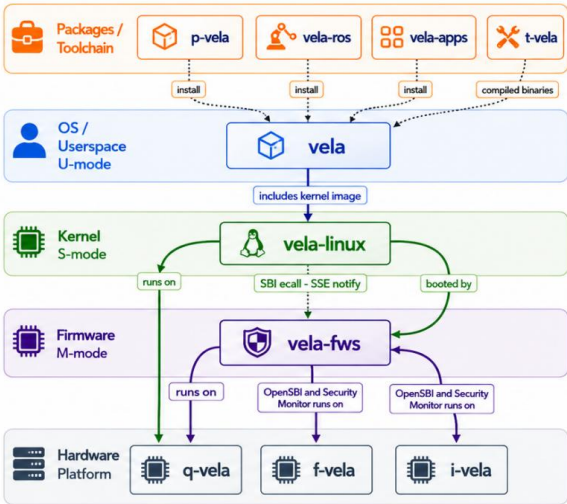
Low Power
Maximize energy efficiency
through power management
and thermal control.**High Performance**
Integrate multiple AI execution
engines and apply multimodal
scheduling for performance
optimization.**Software Defined
Robotics**
Intelligent, adaptable robots
powered by open, software-
defined technology.**High Reliability**
Support a Trusted Execution
Environment (TEE) for
confidential computing.**High Availability**
Implement RAS (Error-corrected
Register Interface) for system error prediction,
recovery, and continuous operation.

RISC-V Vela Software Stack & Repositories

Repository Overview

Repository	Role	Layer	Language	Status
vela-apps	IF, Runtime/SDK	System SW	C/C++	✓ Public
vela-ros	ROS2	System SW	C	✓ Public
vela	Linux Distribution	OS	Shell	✓ Public
vela-linux	Linux Kernel + Driver	OS	C	✓ Public
p-vela	Power Management	System SW	Python	🕒 Planned
t-vela	Cross AI Compiler	Tools	C++	🕒 Planned
vela-fws	Bootloader/Monitor	FW	C	✓ Public
q-vela	RISC-V SW Emulation-QEMU	HW	C	✓ Public
f-vela	RISC-V FPGA	HW	Chisel, Scala, C++	🕒 Planned
i-vela	Enhanced RISC-V FPGA	HW	Chisel, Scala, C++	🕒 Planned

Repository Relationships



Application Domains

Collaborative Robot

Drone · UAM

Smart Factory

AMR

Secure AI Device

Edge AI